

SERVICE RUNBOOK

Database Cluster

service: database-cluster

TIER 0

ON-CALL

1 Service Overview

`database-cluster` is the shared backend data store used by `payments-gateway` , `auth-service` , and other core services. It runs a **primary-replica topology** with automatic failover.

IMPACT This is **shared infrastructure** — a failure here cascades to every dependent tier-1 service at once. Treat any write unavailability as a multi-service incident.

2 Common Failure Patterns

Connection saturation during batch jobs is a recurring issue when batch processes open per-record connections instead of using a shared pool.

Disk usage growth is commonly tied to write-ahead log accumulation if automated log cleanup has been disabled by a configuration change. Schema migrations run without online / non-blocking tooling can cause **extended table locks** and write timeouts across dependent services.

Unexpected failovers are sometimes triggered by brief transient network blips rather than genuine node failure; review health check sensitivity if this recurs without a clear root cause.

SYMPTOM	LIKELY CAUSE / FIRST CHECK
Write failures during batch / scheduled job	Connection pool exhaustion from batch job — check per-record connection pattern
Disk usage alert nearing capacity	WAL log accumulation — check automated cleanup job status

SYMPTOM	LIKELY CAUSE / FIRST CHECK
Write timeouts across multiple services during deploy	Blocking schema migration — check for non-online ALTER TABLE
Unexpected failover, no clear cause	Possible transient network blip — review health check sensitivity

CRITICAL / WRITE IMPACT DEGRADATION

3 Diagnostic Steps

- 1

Check **active connection count** against configured max during the incident window.
- 2

Check **disk usage trend** and WAL cleanup job execution history.
- 3

If write timeouts are reported during a deploy, check whether a migration used **blocking DDL**.

4 Escalation Criteria

- ESCALATE IMMEDIATELY FOR:
- Any **failover event**.
 - Any **write unavailability exceeding 60 seconds**.
 - Disk usage **exceeding 90% on any node**.